PRATHAMESH MHAPSEKAR

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EDUCATION

Degree	Institute / School	Year	Score
Master of Science (Computer Science)	Viterbi School of Engineering (University of Southern California)	Dec 2024	3.85/4
Bachelor of Engg. (Information Technology)	D. J. Sanghvi College of Engineering (University of Mumbai)	May 2019	9.28/10

WORK EXPERIENCE

Software Development Engineer II, JPMorgan Chase & Co. **Jan'22 – Nov'22**

- Developed an integration service layer in Java Spring Boot, which processes and **transforms over a million transaction-status messages** received per day via different IBM MQs & Kafka topics.
- Published messages to firm-wide AMPS and Kafka topics with an **improved TPS of 90 msg/sec by leveraging async computation** in Java using CompletableFuture class, committing to the downstream warehousing system's SLAs.
- Designed and implemented a Kafka producer service that monitors the health API on a regular basis and failovers from the primary Kafka cluster to the replicated cluster hosted on the secondary data center in the event of a health warning, **resulting in zero production incidents related to data center failure.**
- Responsible for **code reviews** to ensure good code quality and **incorporation of design patterns** defined by the core team.

Software Development Engineer I, JPMorgan Chase & Co. **Jul'19 - Dec'21**

- Took charge of the design and development of the 'APAC Cross-Border High-Value Payments Real-Time UI Dashboard,' resulting in a **reduction in retrieval time from around 3 hours to 1 minute** for obtaining various insights into payments data.
- Implemented React JS based client-side application leveraging redux for state management and also delivered a **Progressive Web App** (PWA) implementation for **increasing user engagement and enhancing user experience**.
- Wrote unit test cases for the React JS application using Jest and Enzyme and integrated them with the **Jenkins-based CICD** pipeline to deploy it on the cloud-foundry-based PAAS cloud platform, **cutting deployment time by 75%.**
- Ensured high scalability and reusability by developing a single **configuration-driven UI** Screen to render multiple markets as per their configurations stored in the DB tables thus reducing development time and helping to migrate 27 markets within a year.

PROJECTS

APAC Cross-Border High-Value Payments Real-Time Dashboard and Warehouse Integration **Aug'20 – July'21**

- Designed a **real-time payments** dashboard to **track** and get insights into the Cross-Border High-Value Payments worth **$150bn/day.**
- Helped business operations meet deadlines and SLAs by **enabling real-time monitoring of payments** processed on an IBM mainframe by **recording payment status change** messages received over MQ into DB2 DB and displaying the data on a **reactJs-based dashboard**.
- **Combined multiple downstream feeds** for over a **million transaction-status messages** received via multiple MQs into one JSON payload following the warehousing system's JSON schema and publishing the payload over the Kafka topic on the receipt of all feeds.

Document analysis and offline character recognition for handwritten Devanagari script **Aug'18 – Jul'19**

- **Conducted experiments on four deep CNN architectures** – ResNet, AlexNet, VGG, and Inception for predicting the main body of a Devanagari character which concluded that the Resnet-50 architecture provides the highest accuracy.
- Refactored the code to **enable multi-processing** and deployed it on GCP's IAAS Compute Engine, leveraging its multiple cores.

HONORS AND AWARDS

- Won the **TCS Best Student Project Award '19** for "Document analysis and offline handwritten Devanagari script recognition" project.
- Won the **Best Project Award at the Dipex 2019** – state level project competition under the computer science category.
- Among the top 9 finalists of **AICTE's Chhatra Vishwakarma Awards '17**, represented Maharashtra at the AICTE, New Delhi.
- Secured the runner-up position at the national level **Smart India Hackathon 2018** for the project "Helping Hands".
- Runner up of the **Smart Mobility Hackathon 2019** – national level hackathon organized by HERE Maps.
- Stood second at **VJTI's Technovanza Android Hackathon** organized by m-Indicator and VJTI.
- Awarded the **DPS – Quarterly Excellence Award** for the project "APAC High-Value Payments Real-Time UI Dashboard" at **JPMC**.

PUBLICATIONS

Implementation of Residual Network (ResNet) for Devanagari Handwritten Character Recognition in Advanced Computing Technologies and Applications – Proceedings of 2nd – ICACTA 2020 (https://doi.org/10.1007/978-981-15-3242-9_14).

SKILLS

Languages: C, Java, JavaScript, Python.
Web Technologies: React JS, Springboot, Django
Database Technologies: MySQL, Oracle, IBM DB2
Other Technologies: Git, Rest API, Unix, Apache Kafka, Kubernetes, Eureka